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The Trade Imbalance Network and Currency Returns

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1 Big picture

- **Question.** How do global trade imbalances shape currency risk premia when international financial markets are imperfect and financiers have limited risk-bearing capacity?
- **Core idea.** A currency’s risk premium is not determined only by its own external imbalance. It also depends on where that country sits in the directed global trade-imbalance network.
- **Theory.** The paper extends Gabaix and Maggiori’s limited-commitment intermediary model to a multi-country network. Financiers intermediate country deficits and surpluses, and their outside options depend on the complexity of their global balance sheets.
- **Network object.** The global trade-imbalance network is a directed weighted graph. An edge from country i to country j records the dollar value of country j ’s net imports from country i .
- **Centrality.** The relevant centrality is based on a Leontief inverse. This captures both direct imbalance exposure and indirect exposure through neighbors and neighbors’ neighbors.
- **Empirical characteristic.** The authors construct a Centrality-Based Characteristic, CBC, from the calibrated network centrality and the variance–covariance matrix of currency returns.
- **Main empirical finding.** Sorting currencies on CBC generates a strong long–short currency strategy. The high-CBC minus low-CBC portfolio earns about 3.2–3.6 percent annualized excess return and Sharpe ratios around 0.54–0.65.
- **Novelty.** The CBC strategy is not subsumed by standard currency characteristics such as carry, global imbalances, total trade network centrality, or intermediary asset-pricing factors.
- **Mechanism.** The most important component is not the country’s own total imbalance. It is the neighborhood component: how the country is connected to other imbalanced countries in the network.
- **Risk-bearing evidence.** CBC returns comove with proxies for intermediary risk-bearing capacity such as TED, VIX, VXY, global bid–ask spreads, and commercial paper outstanding.
- **Counterfactual use.** The framework can evaluate how international events, such as the US–China trade war or sanctions on Russia, alter global currency premia through network spillovers.
- **Takeaway.** Currency premia reflect a network externality in trade imbalances. A multi-country incomplete-markets model uncovers risk-premium information that bilateral or one-country imbalance measures miss.

2 Data and measurement

2.1 Trade imbalance network

The paper builds a directed weighted network using annual bilateral import and export data from UN Comtrade for 1995–2021. Let A_t be the adjacency matrix. For $i \neq j$,

$$\{A_t\}_{ij} > 0 \tag{1}$$

means that country j is a net importer from country i . Equivalently, the edge points toward the debtor or deficit country.

Interpretation: The direction of the network is essential. Total trade intensity is undirected, but trade deficits are directional. A country can be central because many other imbalances pass through it, even if its own bilateral imbalance with one country is not extreme.

2.2 Currency returns

The empirical analysis uses monthly currency excess returns for 41 currencies from 2003 to 2021. The underlying currency data include spot and forward exchange rates from Thomson Reuters Datastream. For the euro area, the paper aggregates trade information across regions adopting the euro.

For currency i , the excess return is conceptually

$$rx_{i,t+1} \approx f_{i,t} - s_{i,t+1}, \quad (2)$$

where $f_{i,t}$ is the log forward rate and $s_{i,t+1}$ is the future log spot exchange rate, expressed for a US investor.

Interpretation: Monthly excess returns are the pricing outcomes. The sorting characteristics are based on annual network and return-covariance information, so portfolios are rebalanced yearly.

2.3 Capital-flow and alternative networks

To test whether the trade-imbalance network is special, the paper constructs alternative CBC measures using:

- the CPIS portfolio-investment network;
- a residency-adjusted CPIS network;
- the FDI network;
- Richmond’s total trade network;
- total trade imbalance and variance–covariance-weighted imbalance measures.

Interpretation: The main empirical comparison is not just “network versus no network.” It is whether the directed trade-imbalance network outperforms plausible alternative networks and classic currency characteristics.

2.4 Risk-bearing-capacity proxies

The paper studies whether CBC is linked to intermediaries’ risk-bearing capacity using proxies such as:

- TED spread;
- VIX;
- VXY-G7 and VXY-EM currency volatility indices;
- financial commercial paper outstanding;
- global FX bid–ask spread;
- net sovereign debt holdings by domestic versus foreign investors.

Interpretation: If CBC captures the risk borne by financial intermediaries, then CBC strategy returns should be high when risk-bearing capacity is scarce and should correlate with these stress or liquidity proxies.

3 Model and empirical construction

3.1 Market-clearing in the imbalance network

Let q_i be the dollar value of country i ’s bonds held by financiers, and let ℓ be an $n \times 1$ vector of ones. In equilibrium, the bonds absorbed by financiers satisfy

$$(A_0 - A_0^\top)\ell + q = 0. \quad (5)$$

Interpretation: The vector $(A_0 - A_0^\top)\ell$ is the total net imbalance by country. Financiers must absorb these imbalances through long and short positions in country bonds and currencies.

3.2 Limited-commitment financiers and the pricing equation

The financiers choose bond positions subject to a quadratic outside-option constraint. The central pricing equation is

$$E(\bar{x}_1 - x_0) = \Gamma q. \quad (10)$$

Here $\bar{x}_1$ is the interest-adjusted future exchange-rate vector, x_0 is the current exchange-rate vector, q is the bond-position vector, and Γ measures financiers' balance-sheet risk and complexity.

Interpretation: Currency premia are high when intermediaries must absorb positions that are costly relative to their risk-bearing capacity. The matrix Γ translates network positions into expected appreciation.

3.3 Leontief inverse and centrality

The model specifies

$$\Gamma = [wI + (1 - w)V] \Omega, \quad \Omega = (I - \alpha A_0^\top)^{-1}, \quad (9)$$

where V is the variance-covariance matrix of currency returns, and $w, \alpha \in [0, 1]$.

The Leontief inverse can be expanded as

$$\Omega = (I - \alpha A_0^\top)^{-1} = \sum_{k=0}^{\infty} (\alpha A_0^\top)^k. \quad (19)$$

Interpretation: The first term captures own imbalance. The next term captures direct neighbors. Higher-order terms capture indirect paths through the network. This is why a currency's premium can depend on distant countries' imbalances.

3.4 Centrality-Based Characteristic

The empirical version of Γ is

$$\Gamma(V, w, A_0, \alpha) = [wI + (1 - w)V] (I - \alpha A_0^\top)^{-1}. \quad (22)$$

The CBC of a currency is the model-implied currency premium generated by combining Γ with the imbalance vector q .

Interpretation: CBC is a theory-guided characteristic. It is not just a descriptive network statistic: it is designed to approximate the currency risk premium implied by intermediaries' network-constrained risk-bearing problem.

3.5 Total, individual, and neighborhood components

The difference in premia between currencies i and j can be decomposed into three pieces:

$$\begin{aligned} rp_i - rp_j &= (\Gamma_{ii} - \Gamma_{ji}) \Delta q_{ij} \\ &\quad + [(\Gamma_{ii} - \Gamma_{ji}) - (\Gamma_{jj} - \Gamma_{ij})] q_j \\ &\quad + \sum_{k \neq i, j} (\Gamma_{ik} - \Gamma_{jk}) q_k. \end{aligned} \quad (27)$$

Interpretation: The first term is total imbalance. The second is individual importance. The third is neighborhood importance. Empirically, the neighborhood term explains most of the variation.

4 Empirical design

4.1 Portfolio sorts

Currencies are sorted annually into portfolios using lagged CBC and benchmark characteristics. The main long-short strategy buys high-CBC currencies and shorts low-CBC currencies.

Interpretation: A successful sort means that the network characteristic predicts future currency excess returns out of sample. The use of lagged annual characteristics helps avoid look-ahead bias.

4.2 Benchmark characteristics

The benchmarks include:

- total trade network centrality (TTNC);
- total trade imbalance (TImb);
- carry or forward discount (FD);
- global imbalance (GImb);
- variance-weighted trade imbalance;
- the dollar factor (DOL);
- alternative CBCs built from CPIS, FDI, and total trade networks.

Interpretation: These comparisons separate the contribution of the directed imbalance network from standard currency effects and from other possible network structures.

4.3 Spanning tests and asset-pricing tests

The paper regresses the CBC factor on standard currency and intermediary factors, and also regresses those factors on CBC. It then runs Fama-MacBeth cross-sectional asset-pricing tests on 24 test portfolios.

Interpretation: If CBC is a genuine risk factor, it should not be spanned by existing factors, and it should price the cross-section of test assets.

4.4 Mechanism tests

The paper directly examines mechanism evidence by:

- decomposing model-implied premia into total imbalance, individual importance, and neighborhood importance;
- correlating CBC strategy returns with risk-bearing-capacity proxies;
- comparing CBC to domestic-minus-foreign sovereign debt holdings;
- testing whether a complete-markets trade-network model can rationalize the findings;
- running counterfactual event studies for the US-China trade war and Russia sanctions.

Interpretation: The mechanism evidence is important because the strategy’s profitability alone would not prove that limited intermediary risk-bearing capacity is the source of the premium.

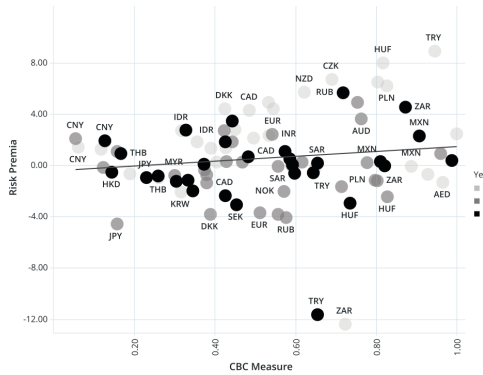
5 Tables and figures

5.1 Figure 1 and Table 1: basic intuition

Read: Figure 1. The scatter plot shows a positive relationship between CBC and average currency risk premia. High-CBC currencies such as TRY, CZK, HUF, ZAR, MXN, and NZD tend to offer high premia, while low-CBC currencies such as JPY, HKD, THB, CNY, and CHF tend to offer low premia.

Read: Table 1. The three-currency example shows how financiers’ long and short positions depend on pairwise deficits and how the Leontief inverse adjusts divertable funds and risk-sharing terms.

Economic message: The economic intuition is visible before the formal tests. Currencies central in the imbalance network must compensate financiers for bearing more complex balance-sheet risk.



g. 1. CBC versus currency risk premia.

Notes: This figure plots the global trade CBC measure of 41 currencies versus average of annualized risk premia. The plots contain averages over three periods, i.e., 2003–2008, 2009–2015, 2016–2021. The trade data are collected from the UN Comtrade Database. The foreign exchange (spot and forward currency) data are collected from Thomson Reuters Datastream. The centrality measure is the *Euro* area, we construct a region by aggregating trade information on countries that adopted the euro in 1999.

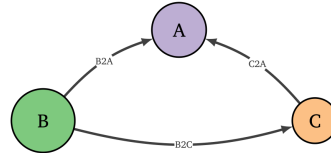
(a) CBC versus currency risk premia

Table 1

A three-currency example with financiers’ long and short positions and their divertable funds.

	Long	Short	Net imbalance q	DB and FB adjusted divertable funds	Risk sharing adjustment
Financier A	DB: B2A FB: C2A	FB: B2A FB: C2A	B2A + C2A	$\underbrace{f_{AA}(B2A + C2A)}_{\text{long positions}} - \underbrace{(f_{AB}B2A + f_{AC}C2A)}_{\text{short positions}} \underbrace{(B2A + C2A)}_{\eta_A}$	$-B2C(f_{AB} - f_{AC})(B2A + C2A)$
Financier B	FB: B2A FB: B2C	DB: B2A DB: B2C	-B2A - B2C	$\underbrace{(f_{BA}B2A + f_{BC}B2C)}_{\text{long positions}} - \underbrace{f_{BB}(B2A + B2C)}_{\text{short positions}} \underbrace{(-B2A - B2C)}_{\eta_B}$	$-C2A(f_{BA} - f_{BC})(B2A + B2C)$
Financier C	DB: B2C FB: C2A	FB: B2C DB: C2A	B2C - C2A	$\underbrace{(f_{CA}B2C + f_{CB}C2A)}_{\text{long positions}} - \underbrace{(f_{CB}B2C + f_{CC}C2A)}_{\text{short positions}} \underbrace{(B2C - C2A)}_{\eta_C}$	$-B2A(f_{CB} - f_{CA})(B2C - C2A)$

Notes: This table summarizes the balance sheet information of the financiers in a simple three-currency network example. The first and second columns show the long and short positions, respectively. DB denotes Domestic Bonds and FB denotes Foreign Bonds. The third column shows the net imbalance, i.e. the elements in q . The fourth and fifth columns show the two components in the divertable funds: DB and FB adjusted divertable component and risk-sharing component, respectively. The imbalance network is shown in the directed graph below in which the arrow of the edge in each pair points towards the deficit country in that pair:



The trade deficit adjacent matrix is:

$$A = \begin{bmatrix} & A & B & C \\ A & 0 & 0 & 0 \\ B & B2A & 0 & B2C \\ C & C2A & 0 & 0 \end{bmatrix}$$

(b) Three-currency balance-sheet example

5.2 Boxes I–II and Figure 2: why the Leontief inverse reduces risk

Read: Boxes I–II. The paper decomposes divertable funds in the three-currency example and defines the Leontief inverse Ω as a weighted sum of direct and indirect paths.

Read: Figure 2. The bar chart compares the divertable-funds ratio under three settings: diagonal Γ , uniform off-diagonal Γ , and Leontief-inverse Γ . The Leontief inverse delivers the lowest ratio, 0.34, compared with 0.37 and 0.36.

Economic message: The Leontief inverse is not cosmetic. It formalizes how network risk-sharing can reduce balance-sheet risk by accounting for direct and indirect offsetting positions.

$$\begin{aligned}
DF_A &= \left[\underbrace{F_{AA}(B2A + C2A)}_{\text{long positions}} - \underbrace{(F_{AB}B2A + F_{AC}C2A)}_{\text{short positions}} \right] \underbrace{(B2A + C2A)}_{\varphi_A} - \underbrace{B2C(F_{AB} - F_{AC})(B2A + C2A)}_{\text{risk-sharing with B and C}}, \\
DF_B &= \left[\underbrace{(F_{BA}B2A + F_{BC}B2C)}_{\text{long positions}} - \underbrace{F_{BB}(B2A + B2C)}_{\text{short positions}} \right] \underbrace{(-B2A - B2C)}_{\varphi_B} - \underbrace{C2A(F_{BA} - F_{BC})(B2A + B2C)}_{\text{risk-sharing with A and C}}, \\
DF_C &= \left[\underbrace{(F_{CC}B2C + F_{CA}C2A)}_{\text{long positions}} - \underbrace{(F_{CB}B2C + F_{CC}C2A)}_{\text{short positions}} \right] \underbrace{(B2C - C2A)}_{\varphi_C} - \underbrace{B2A(F_{CB} - F_{CA})(B2C - C2A)}_{\text{risk sharing with A and B}}.
\end{aligned}
\tag{15}$$

(15)

(16)

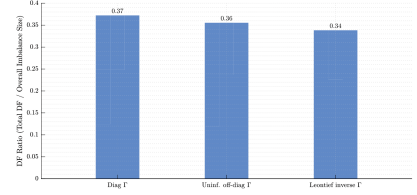
(17)

Box I.

$$\Omega_{ij} = 1_{\{i=j\}} + a \left\{ A_0^1 \right\}_{i,j} + a^2 \sum_{r=1}^n \left\{ A_0^1 \right\}_{i,r} \left\{ A_0^1 \right\}_{r,j} + a^3 \sum_{r=1}^n \left\{ \left(A_0^1 \right)^2 \right\}_{i,r} \left\{ A_0^1 \right\}_{r,j} + \dots,
\tag{18}$$

Box II.

(a) Divertable funds and Leontief inverse



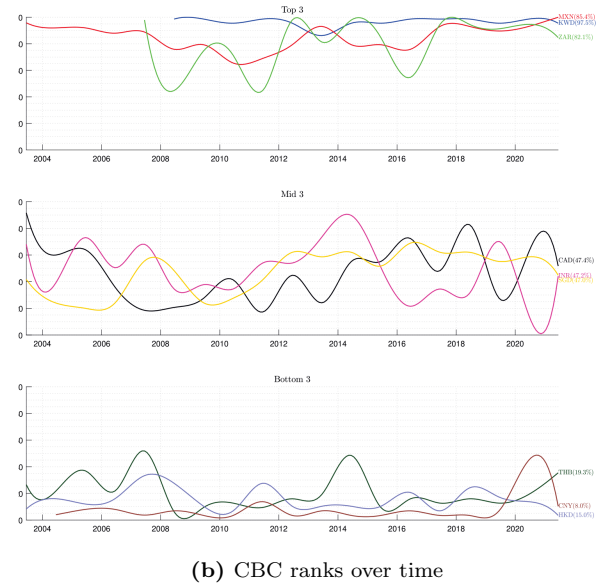
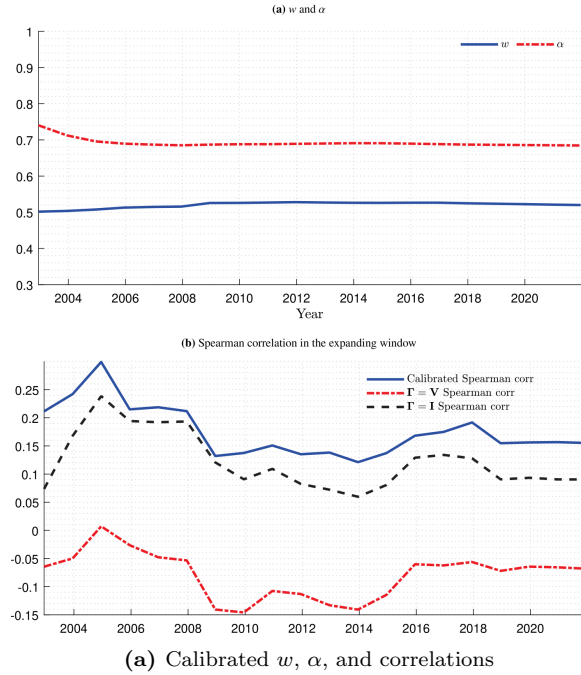
(b) Risk reduction effect

5.3 Figures 3–4: calibration and relative CBC ranks

Read: Figure 3. The calibrated w is stable around 0.5, and α is stable around 0.7. The calibrated Γ produces stronger positive rank correlations with realized currency premia than the special cases $\Gamma = V$ or $\Gamma = I$.

Read: Figure 4. The top CBC currencies are persistent but not fixed. Top-ranked currencies include net importers such as MXN, KWD, and ZAR, while low-ranked currencies include CNY, HKD, and THB.

Economic message: The calibration confirms that both the return covariance matrix and the imbalance network matter. The ranks show that the network characteristic is persistent but still time-varying.



5.4 Table 2 and Figure 5: portfolio sorting performance

Read: Table 2. The CBC updated strategy earns a long–short return of 3.21 percent and Sharpe ratio of 0.54. The static CBC strategy earns 3.58 percent and Sharpe ratio of 0.65. These Sharpe ratios exceed those of standard benchmarks in the table.

Read: Figure 5. Cumulative returns show that CBC dominates after the global financial crisis. Several characteristics perform before 2008, but only CBC continues to trend strongly afterward.

Economic message: The CBC strategy is not just another carry-style sort. It produces sizable out-of-sample currency premia and persistent cumulative performance.

Sorting Var.	G1	G2	G3	G4	G4 - G1				
					Return	Sharpe	FD	SR	SR Return
CBC (updated)		-0.06	0.13	0.20	3.14	0.54	2.54	0.67	0.21
CBC (static)		-0.33	-0.05	0.50	3.25	0.65	2.73	0.85	0.24
Panel A	<i>TTNC</i>	0.74	-0.54	1.30	1.78	0.23	3.04	-2.00	-1.92
	<i>Tmb</i>	0.37	0.37	1.71	0.76	0.08	2.05	-1.65	-4.18
	<i>FD</i>	-0.12	0.49	1.41	1.49	0.21	7.01	-5.40	-3.36
	<i>Gmb</i>	-1.00	0.96	1.72	1.01	0.45	1.41	0.60	0.30
	V-weighted <i>Tmb</i>	0.21	-0.17	1.69	1.44	0.13	1.15	0.08	0.06
	<i>DOL</i>	-	-	-	0.83	0.13	1.39	-0.56	-0.67
Panel B	<i>CPIS</i>	-0.74	-0.33	-1.68	-0.79	-0.01	1.66	-1.72	30.21
	<i>CPIS_{FD}</i>	-0.56	-0.40	-1.80	-0.77	-0.02	1.64	-1.85	8.80
	<i>CPIS_{FD}</i>	-0.62	-0.36	-1.80	-0.73	-0.01	1.71	-1.82	15.75
	<i>FDI</i>	-0.41	-0.50	-2.05	-0.53	-0.01	1.96	-2.08	17.65
	<i>RM_{adj}</i>	0.19	-0.15	1.34	1.83	0.18	1.33	0.30	0.18

Note: This table compares the performance of CBC-sorted portfolio with that of various characteristic-sorted portfolios. Panel A includes: total trade network centrality (*TTNC*), trade imbalance (*Tmb*), forward discount (*FD*), and global imbalance (*Gmb*), variance-covariance weighted trade imbalance (V-weighted *Tmb*), as well as the average currency excess return (*DOL*). *TTNC* is proposed by Richmond (2019), *Tmb* is inspired by Gabaix and Maggiori (2015), *FD* is the carry trade characteristic from Lustig et al. (2011), *Gmb* is used by Della Corte et al. (2016), and V-weighted *Tmb* is inspired by Gabaix and Maggiori (2015, OA). The *DOL* is the currency "market" return in dollars available to a U.S. investor (Lustig et al., 2011). Panel B includes alternative CBCs calibrated on other networks: *CPIS*, *CPIS_{FD}*, *CPIS_{FD}*, *FDI*, and *RM_{adj}*. *CPIS* is the network based on the original financial linkages without any restatements reported in Coordinated Portfolio Investment Survey (CPIS). *CPIS_{FD}* denotes the network based on the CPIS financial linkages but reallocate only securities that, under residency, are issued by affiliates located in tax havens. *CPIS_{FD}* refers to the network based on the CPIS financial linkages and reallocate all securities that, under residency, are issued by affiliates from all countries. Detailed restatement procedures can be found in Coppola et al. (2021). *FDI* is based on bilateral foreign direct investment stocks of all sample countries. *RM_{adj}* refers to the network from Richmond (2019), which is based on the adjacency matrix of total trade. For the characteristics in Panel A and *RM_{adj}*, the sample period is from 2003 to 2021, which is the out-of-sample period for *a* and *w* calibration. For the alternative CBCs based on *CPIS*, *CPIS_{FD}*, *CPIS_{FD}*, and *FDI*, the sample period is from 2008 to 2021 due to the data availability. Two sets of results, "updated" and "static", are reported for CBC where CBC (updated) is calculated based on annually updated *a* and *w* while CBC (static) is based on static *a* and *w* calibrated from the initial sample before 2003. The sorted portfolios are rebalanced yearly. All returns and Sharpe ratios are annualized. All returns are in percentage. All characteristics except the alternative CBCs based on *CPIS*, *CPIS_{FD}*, *CPIS_{FD}*, and *FDI* are constructed based on information from the previous calendar year-end. For the alternative CBCs based on *CPIS*, *CPIS_{FD}*, *CPIS_{FD}*, and *FDI*, *a* and *w* are calibrated in sample using their available sample period from 2008 to 2021. Currencies are sorted in ascending order into quantiles (G1 to G4) for all characteristics except *TTNC* and *Gmb*. Currencies are sorted in the descending order of *TTNC*. For *Gmb* the G1 to G4 are formed from a 2×2 conditional double sorting by *s/a* and *i/a* following Della Corte et al. (2016). The last five columns, respectively, report Return (average return), Sharpe, FD (forward discount), SR (spot return), and $\frac{SR}{Return}$ of the factors from longing G4 and shorting G1. ***, **, and * indicate the statistical significance level of 1%, 5%, and 10%, respectively. The significance level for the average return is calculated from *t*-test.

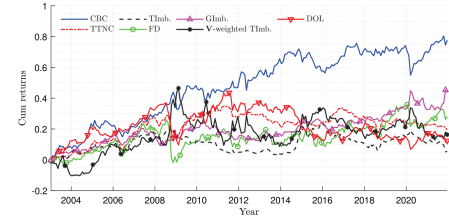


Figure 5: Cumulative returns from 2004 to 2020. The graph shows the cumulative returns of various portfolio sorting strategies. The y-axis represents cumulative returns from -0.2 to 1.0. The x-axis represents the year from 2004 to 2020. The strategies plotted are: CBC (solid blue line), TTNC (dashed red line), Tmb (dotted green line), FD (dash-dot purple line), Gmb (long-dash brown line), V-weighted Tmb (short-dash pink line), and DOL (solid red line with markers). The CBC strategy shows the highest cumulative return, reaching approximately 0.8 by 2020, while most other strategies remain below 0.4.

(a) Portfolio sorting comparison

(b) Cumulative returns

5.5 Tables 3–4: factor spanning and asset pricing

Read: Table 3. Regressing the CBC factor on benchmark factors yields significant annualized alphas, often around 2.8–3.9 percent. Regressing other factors on CBC shows that CBC explains some variation in those factors but does not eliminate all of their alphas.

Read: Table 4. In the Fama–MacBeth tests, $\lambda_{CBC} = 0.344$ with a standard error of 0.024, implying a strongly positive price of CBC risk. The two-factor model with the dollar factor and CBC delivers a high cross-sectional R^2 .

Economic message: CBC is priced in the cross-section and not spanned by standard currency or intermediary factors.

Table 3Time-series variation of *CBC* factor in relation to various factors.

(a) How well various factors explain the <i>CBC</i> factor					
CBC is on LHS	Alpha (%)	T-Alpha	Beta	T-Beta	R^2
<i>TTNC</i>	2.78**	2.43	0.41***	3.51	10.00%
<i>Tlmb</i>	3.18***	2.73	0.06	0.50	0.24%
<i>FD</i>	2.75**	2.24	0.29**	2.23	13.44%
<i>Gtmb</i>	3.35***	3.00	-0.07	-0.40	0.29%
V-weighted <i>Tlmb</i>	3.48***	3.07	-0.22**	-2.38	12.02%
<i>DOL</i>	2.84**	2.42	0.44***	3.33	23.98%
AEM	3.91***	2.82	0.23	1.45	4.14%
HKM	3.04**	2.56	0.11	0.91	0.54%
(b) How well the <i>CBC</i> factor explains various factors					
CBC is on RHS	Alpha (%)	T-Alpha	Beta	T-Beta	R^2
<i>TTNC</i>	0.26	0.26	0.25***	3.80	10.00%
<i>Tlmb</i>	0.27	0.25	0.04	0.49	0.24%
<i>FD</i>	0.10	0.07	0.47***	3.65	13.44%
<i>Gtmb</i>	2.13**	1.97	-0.04	-0.39	0.29%
V-weighted <i>Tlmb</i>	2.97	1.56	-0.55***	-3.05	12.02%
<i>DOL</i>	-0.90	-0.68	0.54***	5.63	23.98%
AEM	-1.52	-1.02	0.18	1.53	4.14%
HKM	1.39	1.43	0.05	0.98	0.54%

Notes: Panel (a) in this table presents the Alpha and Beta coefficients from regressing the *CBC* factor (defined as the G4 - G1 portfolio returns) on various factors. Panel (b) presents the Alpha and Beta coefficients from regressing various factors on the *CBC* factor. The column labeled 'T-Alpha' ('T-Beta') reports the *t*-statistics of Alpha (Beta) estimates based on Newey-West standard errors. Alpha estimates are in annualized percentage. ***, **, and * indicate 1%, 5%, and 10% significance levels, respectively. The sample period is 2003–2021.

(a) Time-series relation between *CBC* and factors**Table 4**

Results from cross-sectional asset pricing tests.

Panel A: Fama-MacBeth regression step one			
	β_{DOL}	β_{CBC}	R^2
CBC 1	0.879***	-0.417***	0.945
Std err	0.014	0.022	
CBC2	1.031***	-0.135***	0.881
Std err	0.026	0.045	
CBC3	1.184***	-0.028	0.932
Std err	0.023	0.027	
CBC4	0.879***	0.583***	0.972
Std err	0.014	0.022	
p-value			
GRS4	0.712		
GRS24	0.167		
$\bar{a}_{abs,4}$	0.035		
$\bar{a}_{abs,24}$	0.041		
Panel B: Fama-MacBeth regression step two			
λ_{DOL}	0.056	R^2	0.753
Std err	0.007	χ^2_{n-k}	
λ_{CBC}	0.344	p-value	0.999
Std err	0.024		
Panel C: Fama-MacBeth regression step two with λ_{CBC} fixed			
λ_{DOL}	0.055	R^2	0.454
Std err	0.006	χ^2_{n-k}	
λ_{CBC}	0.298	p-value	0.999

Note: In this table, we report the results from the Fama-MacBeth cross-sectional regression. The test assets include four portfolios sorted on *CBC* (static), and the five other benchmark characteristics: Total trade network centrality (*TTNC*), proposed by Richmond (2019); trade imbalance (*Tlmb*), defined as total net import (equivalent to setting Γ equal to I); forward discount (*FD*), a common carry trade characteristic in the literature, e.g., Lustig et al. (2011); global imbalances (*Gtmb*), studied by Della Crote et al. (2016); and variance-covariance weighted trade imbalance (V-weighted *Tlmb*), denoted as V_q (equivalent to setting Γ equal to V). *DOL* represents the currency excess return on a portfolio strategy that longs all foreign currencies with equal weights and shorts the domestic currency (USD). Panel A reports the results from the first step of the Fama-MacBeth regression, conducted using the 24 test assets of the six characteristics. 'Std err' denotes the standard error. GRS4 and GRS24 are tests for the four *CBC* and the 24 test assets, respectively, as described by Gibbons et al. (1989) testing whether the intercepts of the regression are jointly different from zero. The p-values are reported. $\bar{a}_{abs,4}$ and $\bar{a}_{abs,24}$ represent the averages of the absolute values of the alphas obtained

(b) Cross-sectional asset-pricing tests

5.6 Box III and Tables 5–6: total imbalance versus neighborhood importance

Read: Box III. The model decomposes pairwise premium differences into total imbalance, individual importance, and neighborhood importance.

Read: Table 5. In pair-year panel regressions, neighborhood importance has the largest standardized coefficient: 0.339 in the univariate regression and 0.326 in the multivariate regression. Total imbalance and individual importance are positive but much smaller.

Read: Table 6. The variance decomposition shows that neighborhood importance accounts for 68.4 percent of the variance in model-implied currency premia, compared with 11.4 percent for total imbalance and 4.9 percent for individual importance.

Economic message: The paper's main mechanism is the neighborhood channel. What matters is not only how imbalanced a country is, but how that imbalance is embedded in the global network.

$$\begin{aligned}
rp_i - rp_j &= \Gamma_{ii}q_i + \sum_{k \neq i} \Gamma_{ik}q_k - \Gamma_{jj}q_j - \sum_{k \neq j} \Gamma_{jk}q_k \\
&= (\Gamma_{ii} - \Gamma_{ji})q_i - (\Gamma_{jj} - \Gamma_{ij})q_j + \sum_{k \neq i \text{ or } j} (\Gamma_{ik} - \Gamma_{jk})q_k \\
&= \underbrace{(\Gamma_{ii} - \Gamma_{ji})\Delta q_{ij}}_{\text{total imbalance}} + \underbrace{[(\Gamma_{ii} - \Gamma_{ji}) - (\Gamma_{jj} - \Gamma_{ij})]q_j}_{\text{individual importance}} + \underbrace{\sum_{k \neq i \text{ or } j} (\Gamma_{ik} - \Gamma_{jk})q_k}_{\text{neighborhood importance}}
\end{aligned} \tag{27}$$

Box III

Figure 6: Box III: decomposition of pairwise currency premia

Table 5

Panel regression results for three components.

	Pair-wise currency premia difference			
Total Imb.	0.142*** (4.62)		0.064** (2.18)	
Indiv.		0.075*** (3.57)	0.074*** (3.73)	
Neighb.			0.339*** (18.06)	0.326*** (16.33)
Observations	7211	7211	7211	7211
Adjusted R ² (%)	2.42	2.16	8.13	8.39
Fixed Effects	Pair & year	Pair & year	Pair & year	Pair & year

Note: This table presents the standardized beta coefficients from panel regressions of the pair-wise currency premia differences on total imbalance (Total Imb.), individual importance (Indiv.) and neighborhood importance (Neighb.), univariate (columns 2 to 4) and multivariate (column 5). The regressions use 7211 pair-year data. *t*-statistics based on robust standard errors are reported in parentheses. Number of observations and adjusted R²'s are reported at the bottom. Both pair-specific and year fixed effects are controlled. The sample period is 2003–2021.

(a) Panel regressions for three components

Table 6

Variance decomposition of model-implied currency premia.

(a) Mean and variance ratio				
	$\Delta rp_{i,j}$	Total Imb.	Indiv.	Neighb.
Mean	-0.014 (0.116)	-0.007 (0.039)	0.005 (0.026)	-0.013 (0.096)
VR	-	0.114 (0.055)	0.049 (0.037)	0.684 (0.131)
(b) Correlation matrix				
	$\Delta rp_{i,j}$	Total Imb.	Indiv.	
Total Imb.	0.568 (0.159)			
Indiv.	0.073 (0.261)	-0.424 (0.305)		
Neighb.	0.958 (0.032)	0.392 (0.159)	-0.007 (0.244)	

Note: Panel (a) presents the sample mean (Mean) of $\Delta rp_{i,j}$, total imbalance (Total Imb.), individual importance (Indiv.) and neighborhood importance (Neighb.), and

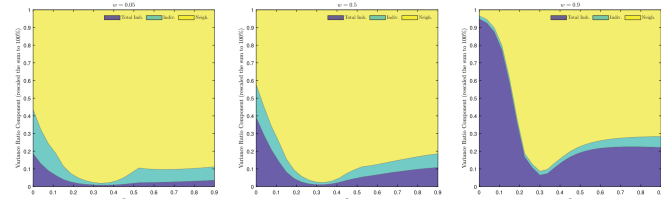
(b) Variance decomposition of model-implied premia

5.7 Figure 6 and Table 7: robustness of the mechanism and risk-bearing capacity

Read: **Figure 6.** Across values of α and w , the neighborhood component remains dominant except near the special case where Γ is close to the identity.

Read: **Table 7.** The CBC strategy has negative correlations with TED, VIX, VXY-G7, VXY-EM, and GBAS, and a positive relation with FCPO. The signs are consistent with high CBC returns when intermediary risk-bearing capacity is weak.

Economic message: The neighborhood mechanism survives parameter variation and is tied to empirical proxies for financing constraints and liquidity conditions.



ance decomposition with varying α and w . Panels in this figure plot the (re-scaled) variance ratios of total imbalance (Total Imb.), individual importance (Indiv.), and neighborhood importance (Neighb.) for α and w from 0 to 0.9 in all three panels. The left panel shows the variance ratios components for $\alpha = 0.05$, the center panel for $\alpha = 0.5$, and the right panel for $\alpha = 0.9$. The results are generated based on the data in the out-of-sample period, from 2003 to 2021. For ease of presentation, the variance ratios are with their relative importance preserved for each combination of α and w .

(a) Variance decomposition as α and w vary

Table 7

Link between imbalance network and risk bearing capacity.

Returns	TED	VIX	VXY-G7	VXY-EM	FCPO	GBAS
CBC strategy	-0.214*** -3.297	-0.345*** -5.530	-0.356*** -5.020	-0.338*** -4.735	0.097 1.459	-0.138** -2.102
	0.001	0.000	0.000	0.000	0.146	0.037
Group 1	-0.061	-0.280***	-0.341***	-0.503***	0.243***	-0.170***
t-statistic	-0.922	-4.388	-4.787	-7.680	3.770	-2.597
p-value	0.357	0.000	0.000	0.000	0.000	0.010
Group 2	-0.150**	-0.372***	-0.454***	-0.562***	0.279***	-0.277***
t-statistic	-2.274	-6.031	-6.713	-8.952	4.369	-4.338
p-value	0.024	0.000	0.000	0.000	0.000	0.000
Group 3	-0.150**	-0.404***	-0.472***	-0.615***	0.298***	-0.212***
t-statistic	-2.275	-6.645	-7.069	-10.275	4.605	-3.254
p-value	0.024	0.000	0.000	0.000	0.000	0.001
Group 4	-0.196***	-0.445***	-0.502***	-0.6081***	0.242***	-0.220***
t-statistic	-3.807	-7.471	-7.658	-10.103	3.744	-3.384
p-value	0.003	0.000	0.000	0.000	0.000	0.001

Note: This table shows correlations between the excess returns from portfolios sorted on CBC and various proxies for risk-bearing capacity, including the TED spread (TED, defined as the interest rate difference between three-month interbank deposit rate and three-month Treasury bill), implied volatility indices for the equity market (VIX) and foreign exchange markets (VXY-G7 for G7 countries and VXY-EM for emerging markets), financial commercial paper outstanding (FCPO), and the global bid-ask spread (GBAS, the global foreign exchange illiquidity measure proposed by Menkhoff et al. 2012a). The table presents the correlations between the excess returns of the CBC strategy (G4 minus G1) and the four groups of portfolios (G1 to G4) sorted in ascending order of CBC, with the proxies for risk bearing capacity. All returns are annualised. All risk bearing capacity proxies are in the form of log differences, except for GBAS, which is in first differences. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

(b) CBC and risk-bearing-capacity proxies

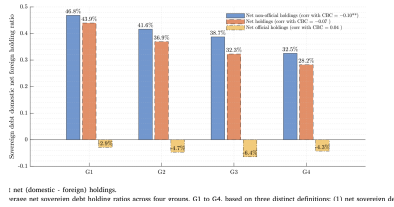
5.8 Figures 7–9: sovereign holdings and counterfactual network shocks

Read: Figure 7. Net sovereign debt non-official holdings decline monotonically from low-CBC to high-CBC groups. The correlation with CBC is negative and statistically significant for non-official holdings.

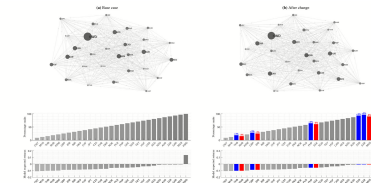
Read: Figure 8. The US–China trade war counterfactual changes not only CNY and USD positions but also the ranks of currencies such as RUB, GBP, IDR, PHP, SEK, ZAR, SAR, and MXN.

Read: Figure 9. The Russia sanctions counterfactual lowers RUB’s rank the most, but also affects currencies such as DKK, THB, and ZAR that are not necessarily direct sanctioning currencies.

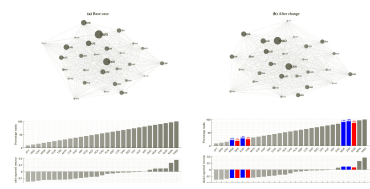
Economic message: Network spillovers are global. A bilateral trade shock can affect many currencies’ premia through indirect imbalance links.



(a) Sovereign debt net holdings



(b) US–China trade-war counterfactual



(c) Russia sanctions counterfactual

6 Short summary

- The paper studies currency returns through a directed global trade-imbalance network.
- It extends Gabaix and Maggiori’s limited-commitment intermediary model to a multi-country setting.
- Financiers absorb global trade imbalances by holding long and short country bond positions.
- Their risk-bearing capacity depends on balance-sheet complexity, represented by a matrix Γ .
- Γ combines currency return covariances with a Leontief inverse of the trade-imbalance network.
- The Leontief inverse captures direct and indirect network links.
- The empirical characteristic CBC is the model-implied currency premium generated by the network.
- Sorting currencies on CBC delivers a high-Sharpe long–short strategy.
- The CBC strategy is not explained by carry, global imbalances, total trade network centrality, the dollar factor, or intermediary factors.
- The neighborhood component explains most variation in model-implied currency premia.
- CBC returns are related to proxies for intermediary risk-bearing capacity.
- The framework can trace how geopolitical or trade shocks alter currency premia through network spillovers.

7 Conclusion

7.1 Core conclusion

Hou, Sarno, and Ye show that currency risk premia are shaped by the global network of trade imbalances. In a multi-country incomplete-markets model, intermediaries demand compensation for absorbing imbalances that

make their balance sheets complex. A Leontief-inverse centrality measure translates this network complexity into a currency characteristic, CBC, that predicts currency excess returns.

7.2 Economic intuition

A country's currency premium rises when financiers must bear costly risk associated with that country's position in the global imbalance network. The country does not need to have the largest direct imbalance. It may have a high premium because its neighbors and indirect network paths make its position central to global risk sharing.

7.3 Takeaway

The paper's main contribution is to turn global trade imbalances into a network asset-pricing characteristic. It shows that currencies are priced not only by interest rates and bilateral balances, but also by where countries sit in a directed network of deficits and surpluses.